

Aliah University

MCA 4th - Semester Examination – 2026

Paper Name: Elective: Internet of Things

Full Marks: 80

Paper Code: MCAPGPE18

Time: 3 hrs

Group A

(Answer all five questions) [5 × 2 = 10]

1. Which protocol is primarily used for lightweight messaging between IoT devices with limited bandwidth?
a) HTTP b) MQTT c) FTP d) SSH CO1, PO1
2. In the context of IoT, what does "Edge Computing" refer to?
a) Centralized cloud storage b) Data processing near the source c) Long-range transmission d) Manual data entry CO1, PO1
3. Which sensor would be most appropriate for measuring the distance of an object in an IoT project?
a) DHT11 b) Ultrasonic Sensor c) LDR d) Potentiometer CO1, PO1
4. What is the main purpose of the "Gateway" in an IoT network?
a) To power sensors b) To bridge different communication protocols c) To act as a physical enclosure d) To display data CO2, PO2
5. Which frequency band is commonly utilized by LoRaWAN for long-range communication?
a) 2.4 GHz b) Sub-GHz (e.g., 868/915 MHz) c) 5 GHz d) 60 GHz CO2, PO2

Group B

(Answer any six questions) [6 × 5 = 30]

6. Define the 3-layer and 5-layer IoT architectures and identify the core responsibilities of each layer. CO2, PO2
7. Explain the significance of the MQTT "Publish/Subscribe" model compared to the traditional HTTP "Request/Response" model. CO3, PO3
8. Discuss the challenges of power management in battery-operated IoT nodes and suggest two energy-saving techniques. CO2, PO2
9. Describe the role of a Microcontroller (e.g., ESP32) in an IoT system versus a Microprocessor (e.g., Raspberry Pi). CO1, PO1
10. Analyze how "Digital Twins" can be used to monitor and simulate physical assets in an Industrial IoT (IIoT) setting. CO3, PO3
11. Explain the concept of "Data Silos" in IoT and how interoperability standards help overcome this issue. CO4, PO4
12. Compare Bluetooth Low Energy (BLE) and ZigBee in terms of range, power consumption, and network topology. CO2, PO2
13. Discuss the ethical concerns regarding data privacy in IoT-enabled smart home environments. CO4, PO4

Group C

(Answer any four questions) [4 × 10 = 40]

14. Analyze the security requirements of IoT devices at the perception layer to prevent physical tampering and side-channel attacks. Critically evaluate the role of Trusted Execution Environments in securing sensitive data within IoT hardware. CO5, PO5
- 5+5

15. Compare the technical specifications and use cases of Narrowband IoT (NB-IoT) against LTE-M for cellular IoT deployments. Discuss how these technologies impact the scalability of smart utility metering in urban and rural areas. CO4, PO4 5+5
16. Analyze the architecture of a Smart Grid system and the role of IoT in balancing energy demand and supply. Critically discuss the challenges of integrating renewable energy sources into the grid using IoT-based monitoring. CO5, PO5 5+5
17. Evaluate the use of Machine Learning algorithms at the "Fog Layer" for real-time anomaly detection in industrial sensor streams. Discuss the trade-offs between local processing latency and the accuracy of cloud-based deep learning models. CO3, PO3 5+5
18. Write an Arduino/ESP8266 script to read data from an LDR and trigger an MQTT message when light falls below a threshold. Develop a pseudocode for an automated irrigation system that activates a pump based on soil moisture and weather forecasts. CO3, PO3 5+5
19. Examine the role of the IPv6 protocol and 6LoWPAN in addressing the massive scaling requirements of the global IoT network. Discuss the header compression techniques used in 6LoWPAN to enable IPv6 over low-power wireless personal area networks. CO4, PO4 5+5